Chapter Six

Comparative Statics of Demand

Properties of Demand Functions

- **Comparative statics is the comparison of two different equilibrium states, before and after a change in some parameter.**
- **How do ordinary demands $x_1^*(p_1,p_2,m)$ and $x_2^*(p_1,p_2,m)$ change as prices p_1, p_2 and m change?**

Properties of Demand Functions

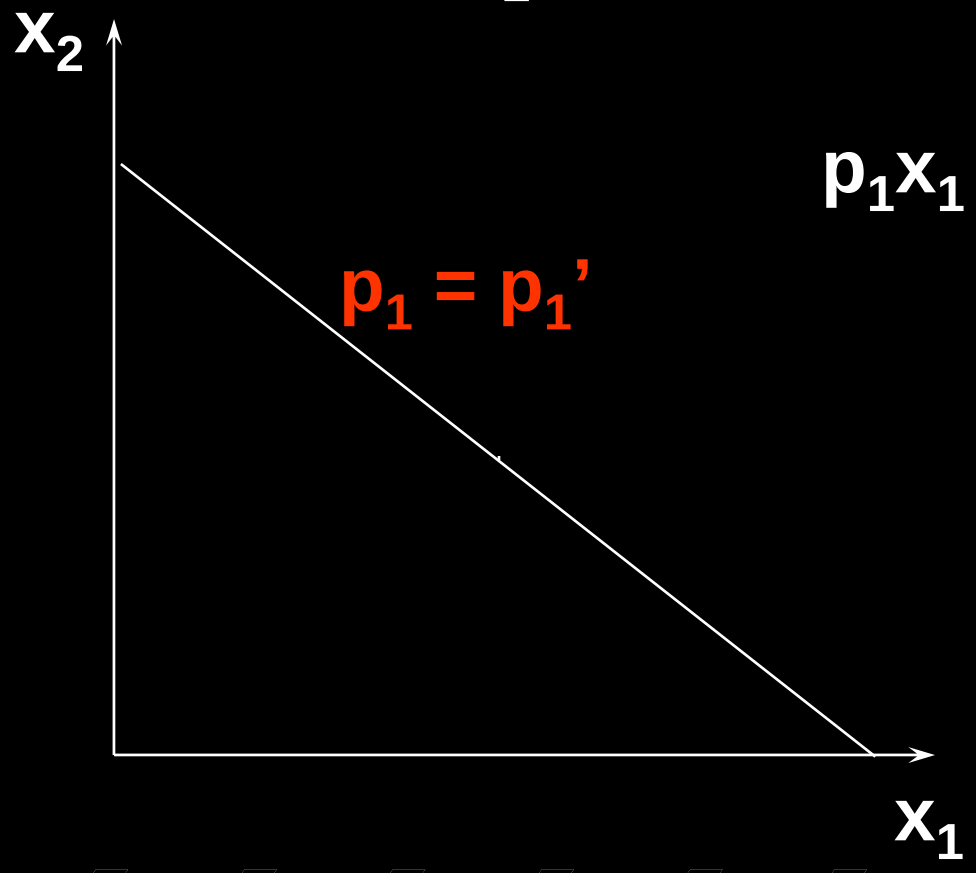
- **Definitions to remember:**
- **Price-related: Ordinary and Giffen goods**
- **Income related: Normal and inferior goods**
- **Cross-price: Gross complements /substitutes**

Own-Price Changes

- How does $x_1^*(p_1,p_2,m)$ change as p_1 changes, holding p_2 and m constant?
- Suppose only p_1 increases, from p_1' to p_1'' and then to p_1'''.

Own-Price Changes

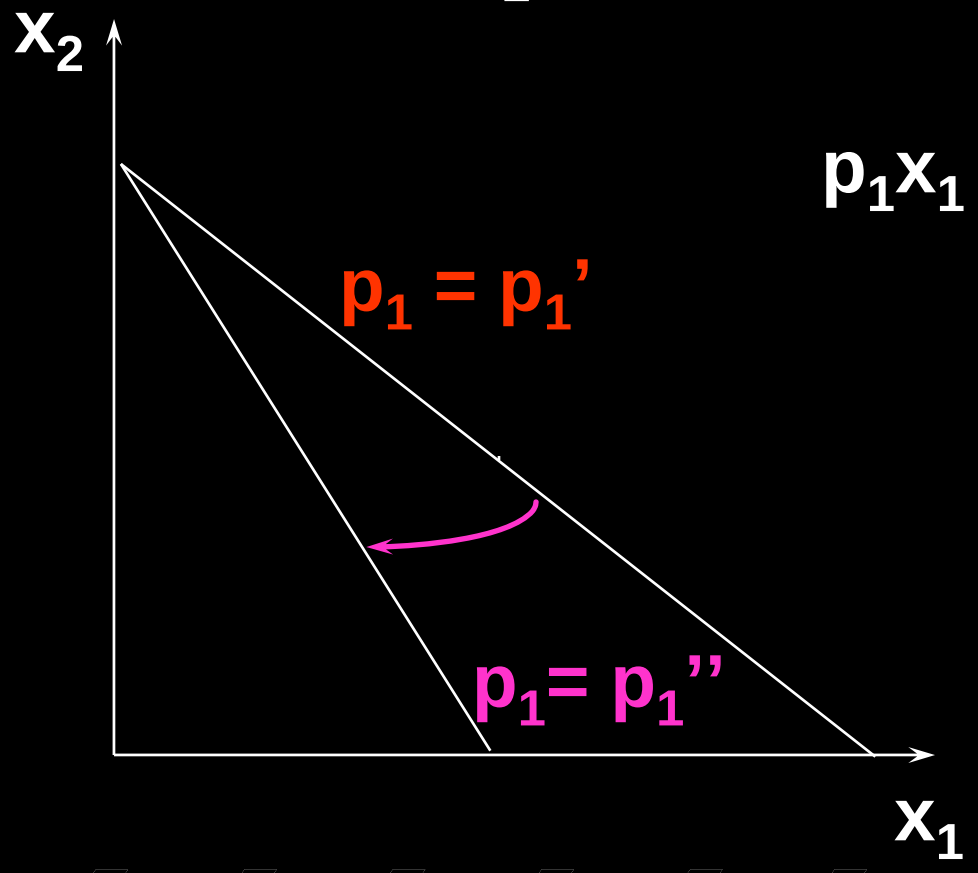
Fixed p_2 and m.



$$p_1x_1 + p_2x_2 = m$$

Own-Price Changes

Fixed p_2 and m.



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Own-Price Changes

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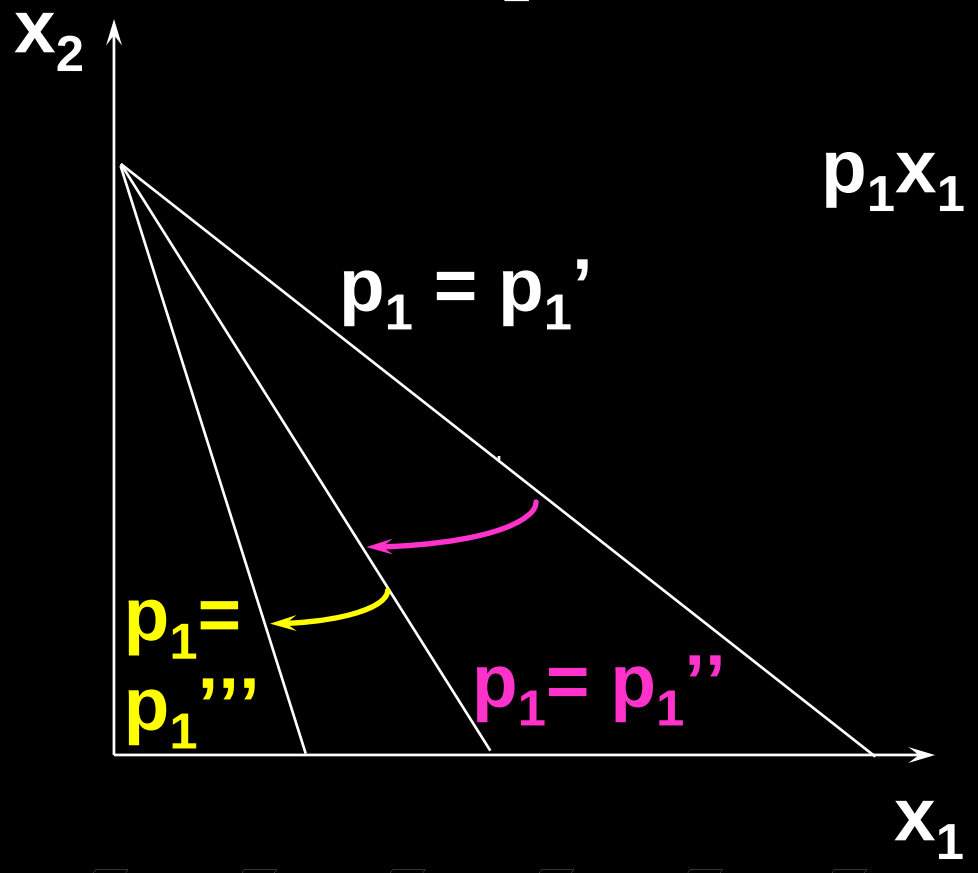
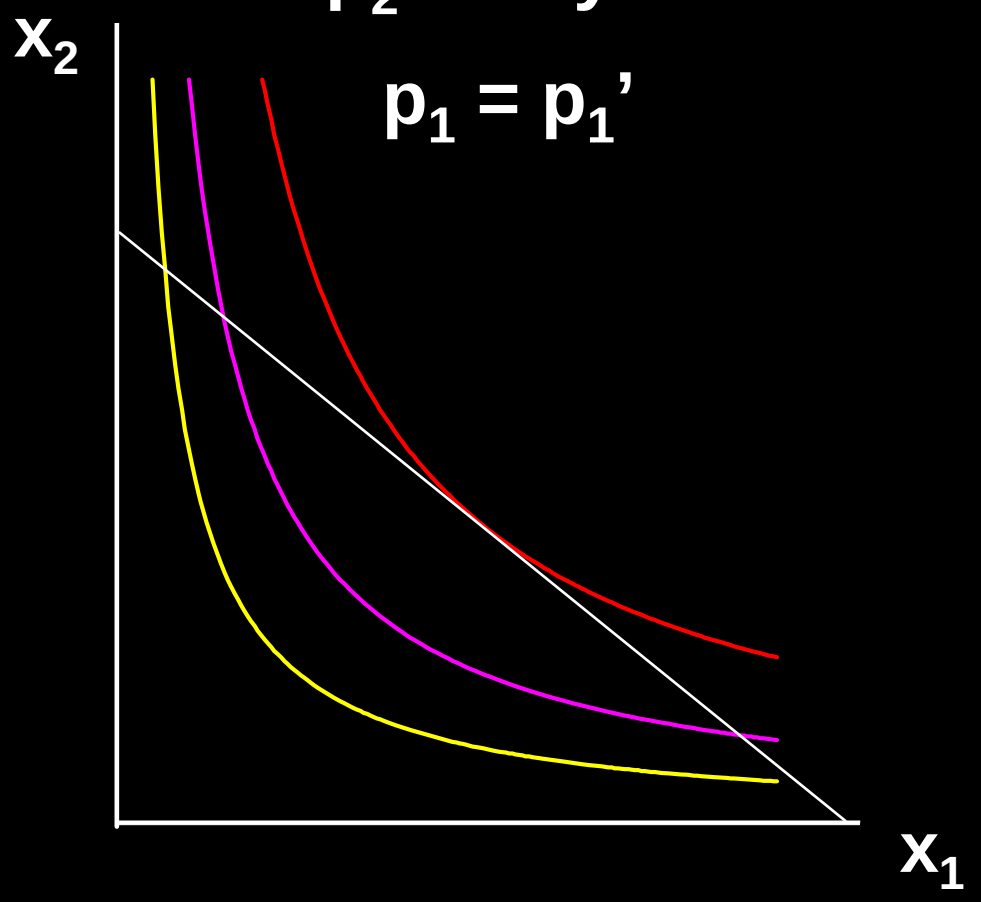


$$p_1x_1 + p_2x_2 = m$$

Own-Price Changes

Fixed p_2 and y.



Own-Price Changes

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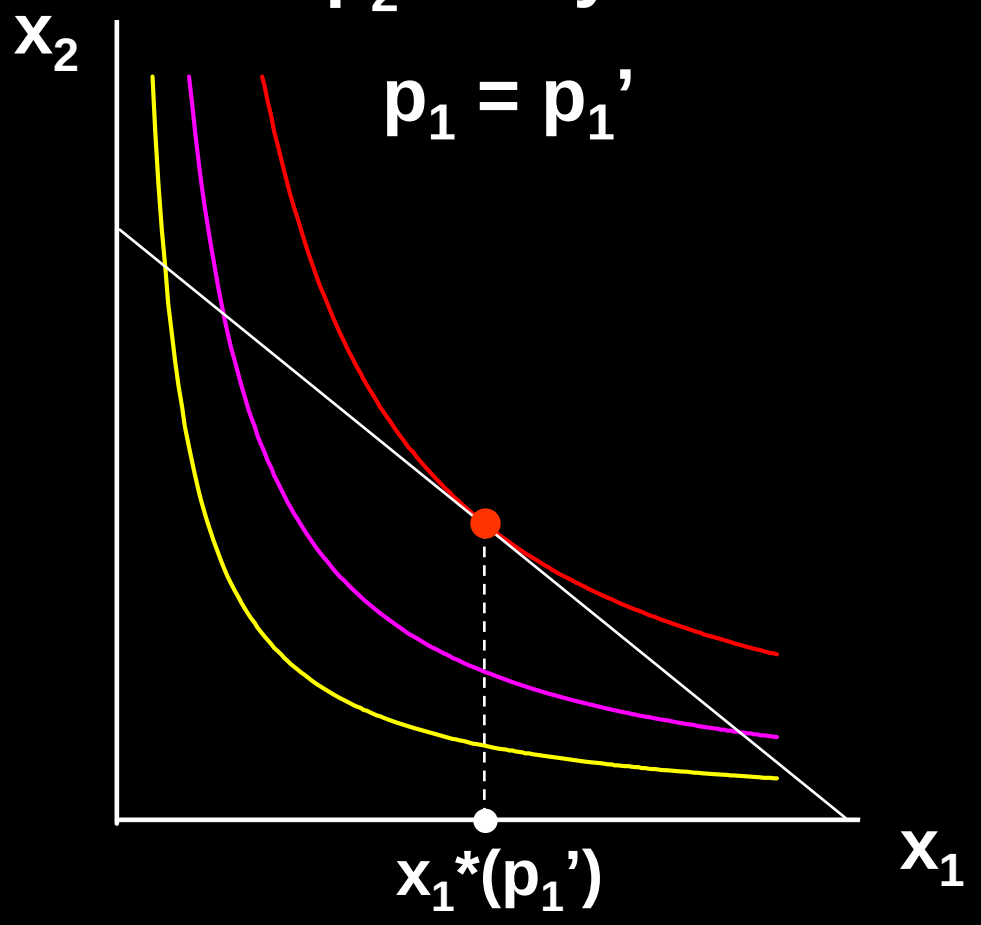
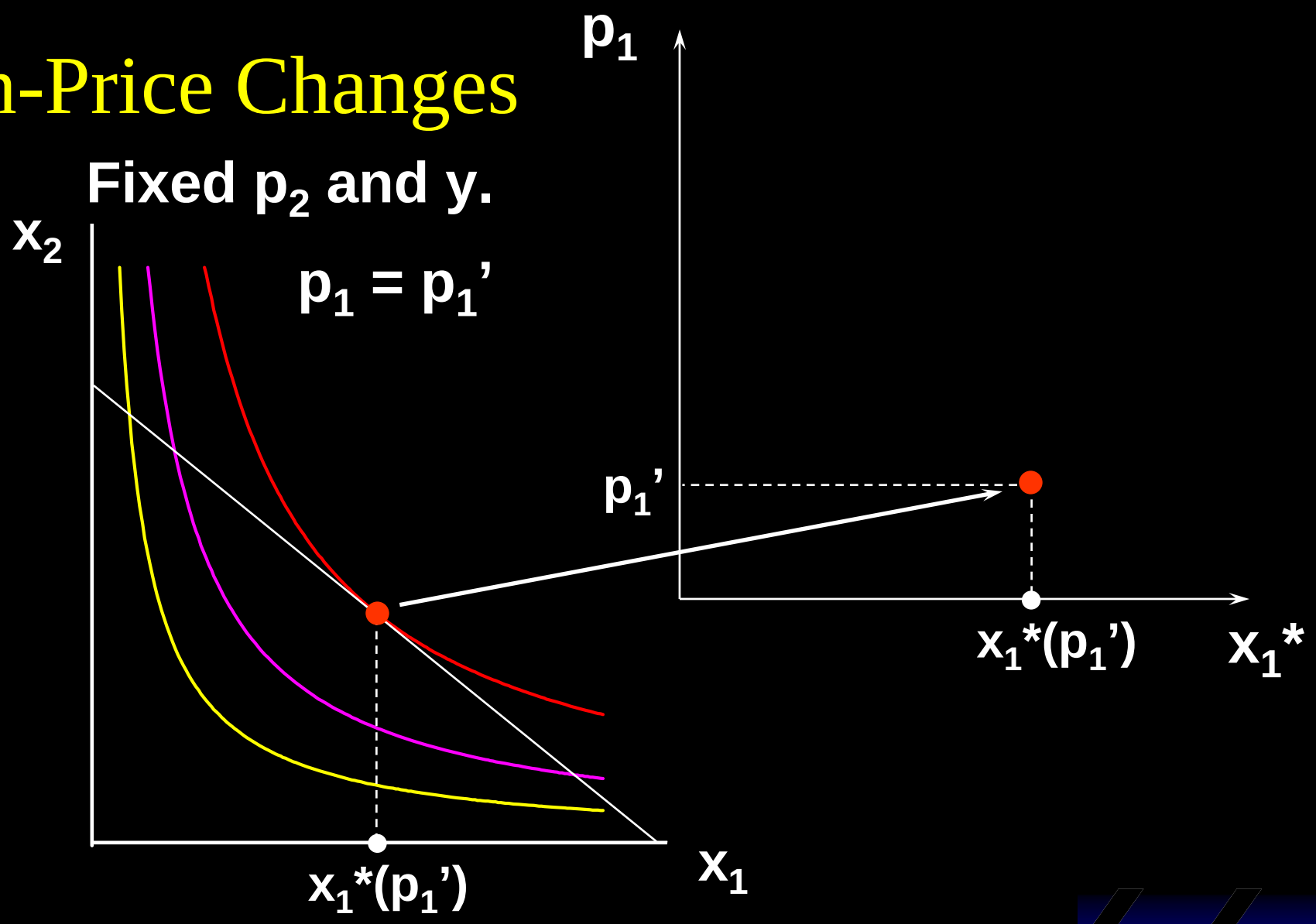
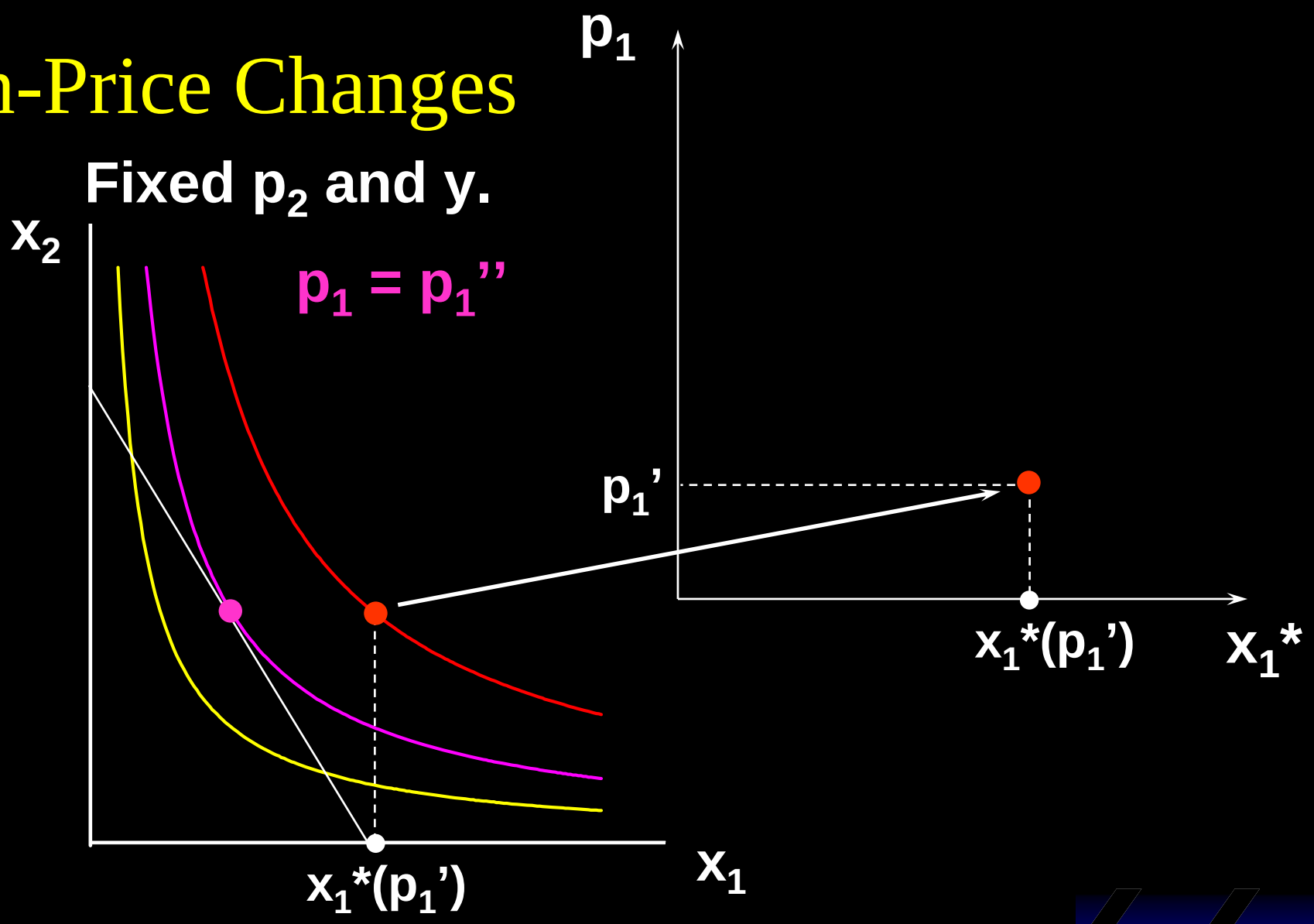


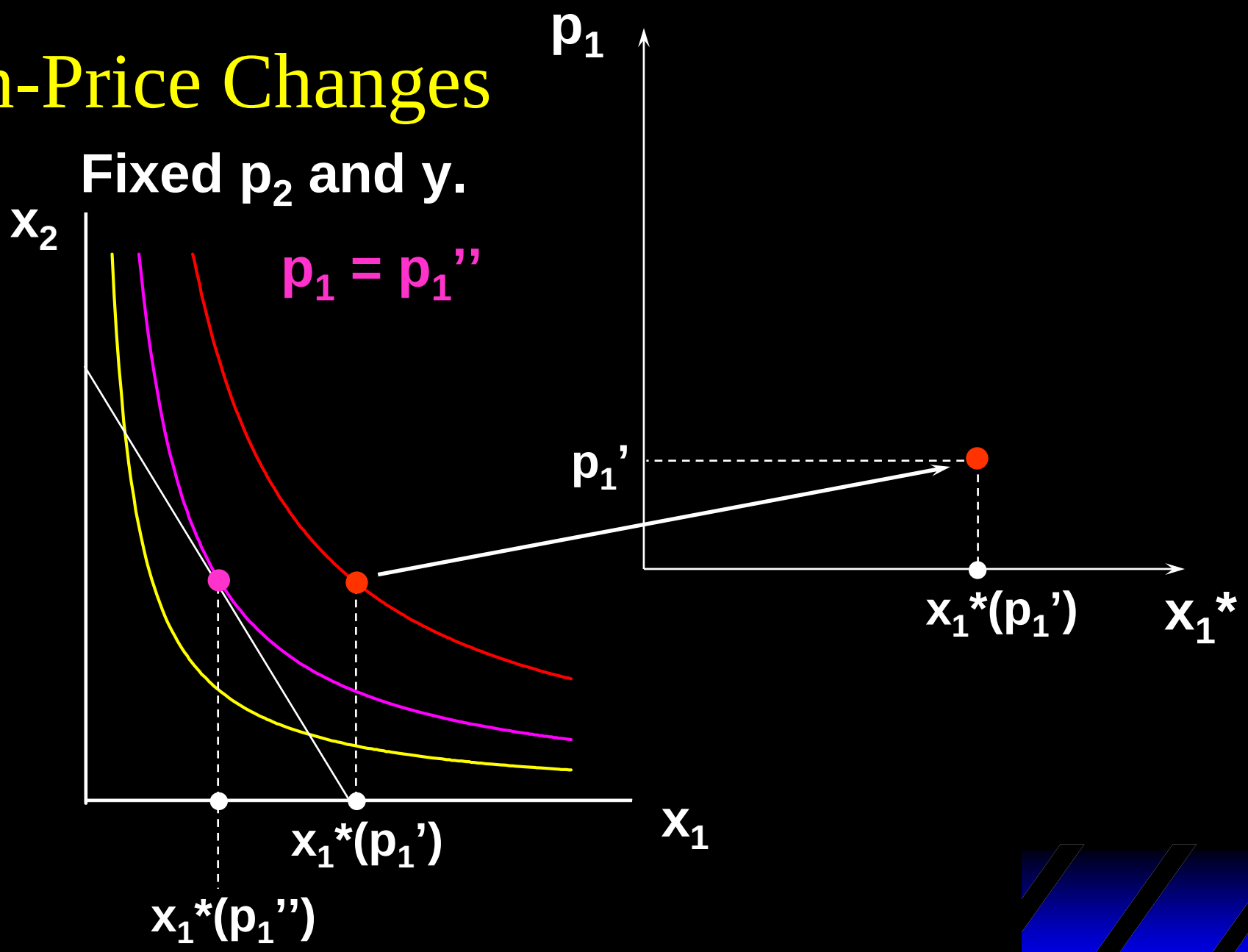
Own-Price Changes



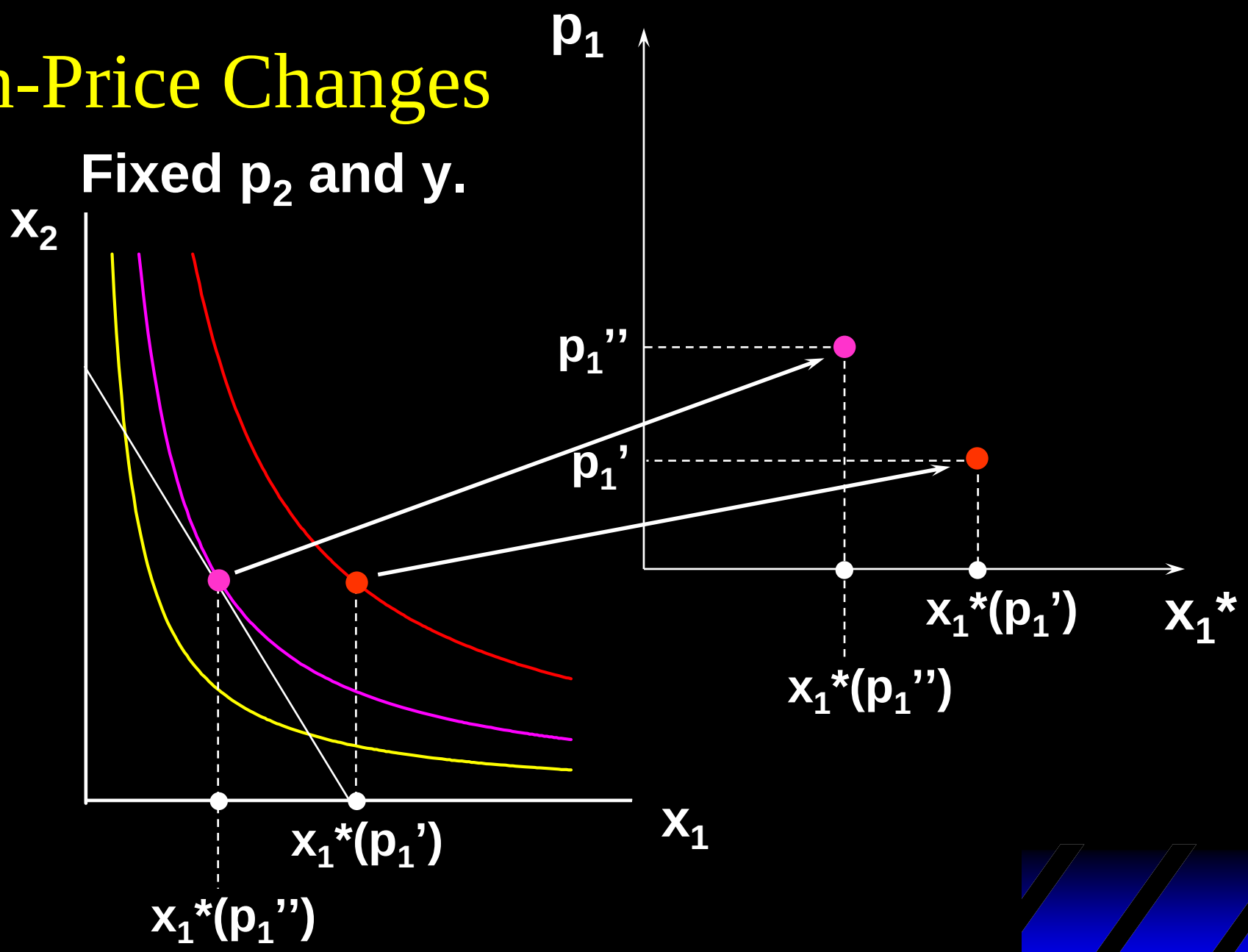
Own-Price Changes



Own-Price Changes



Own-Price Changes



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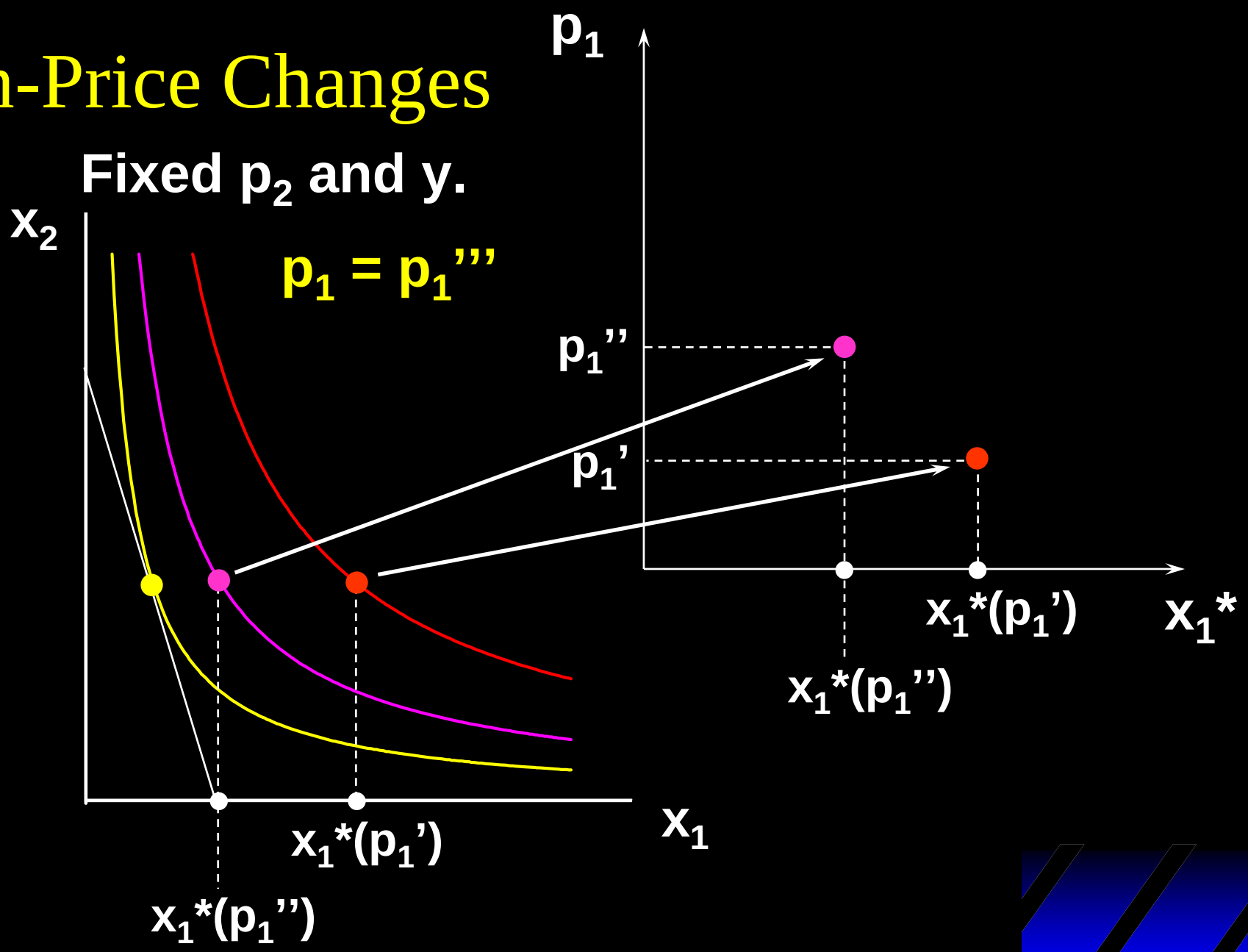
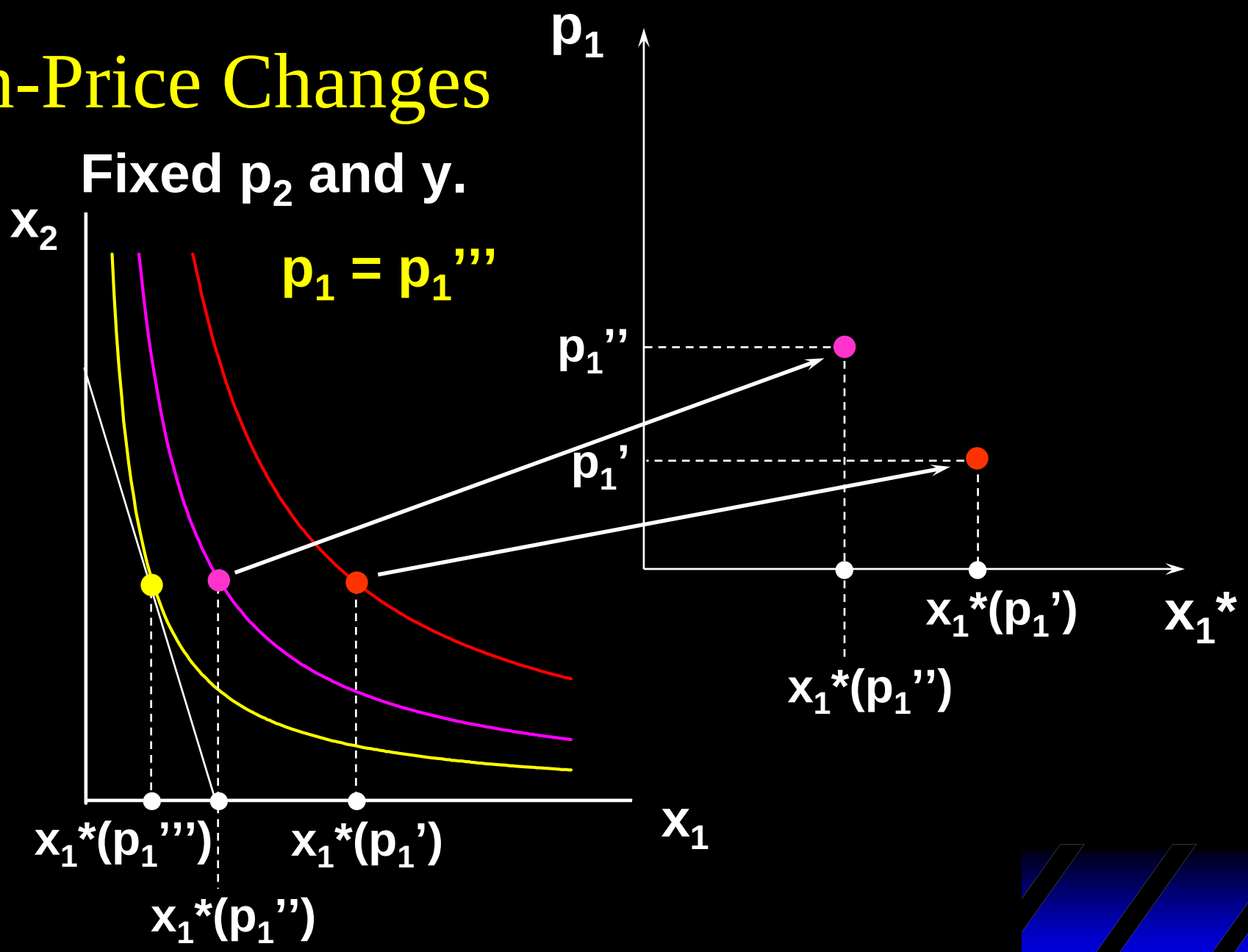
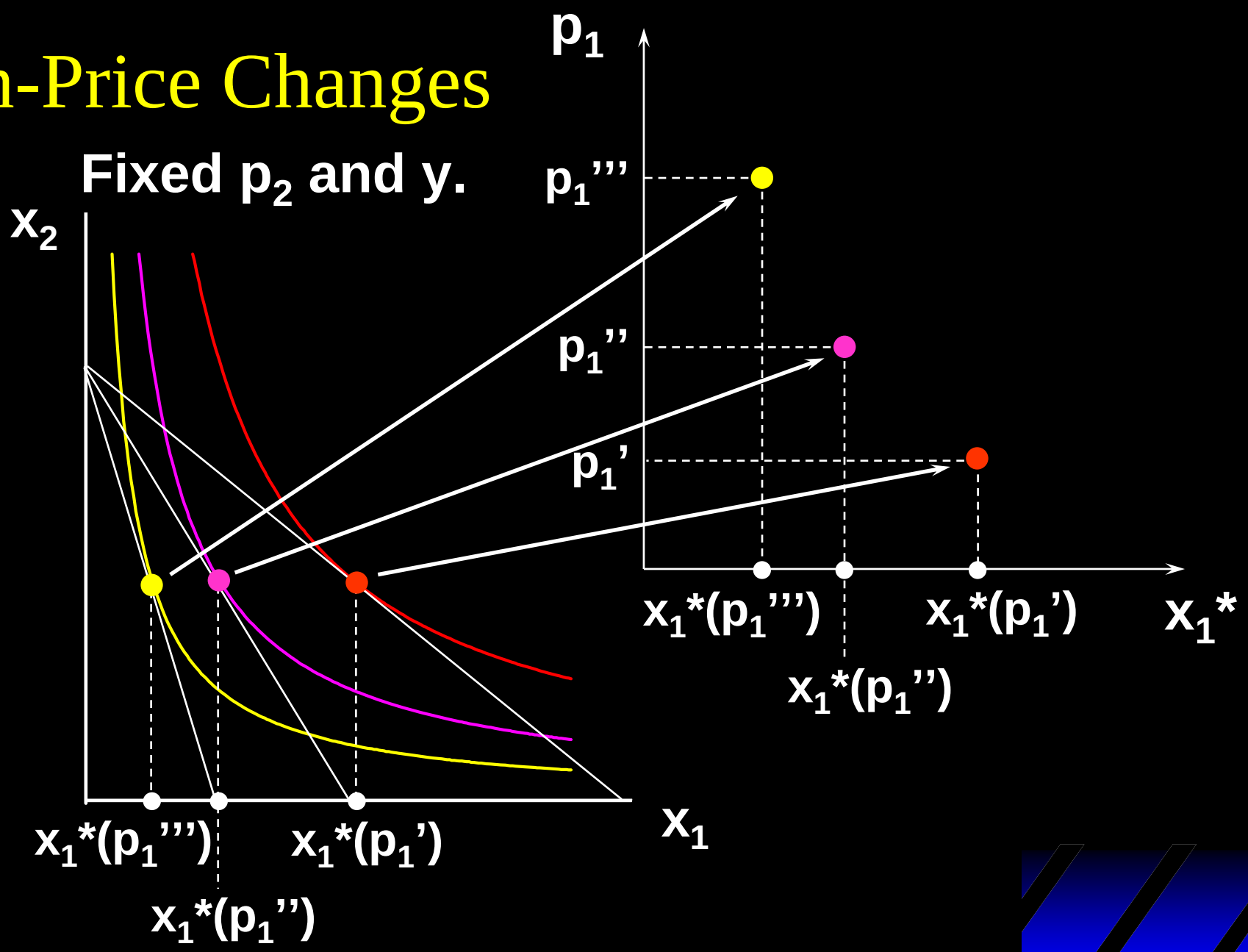


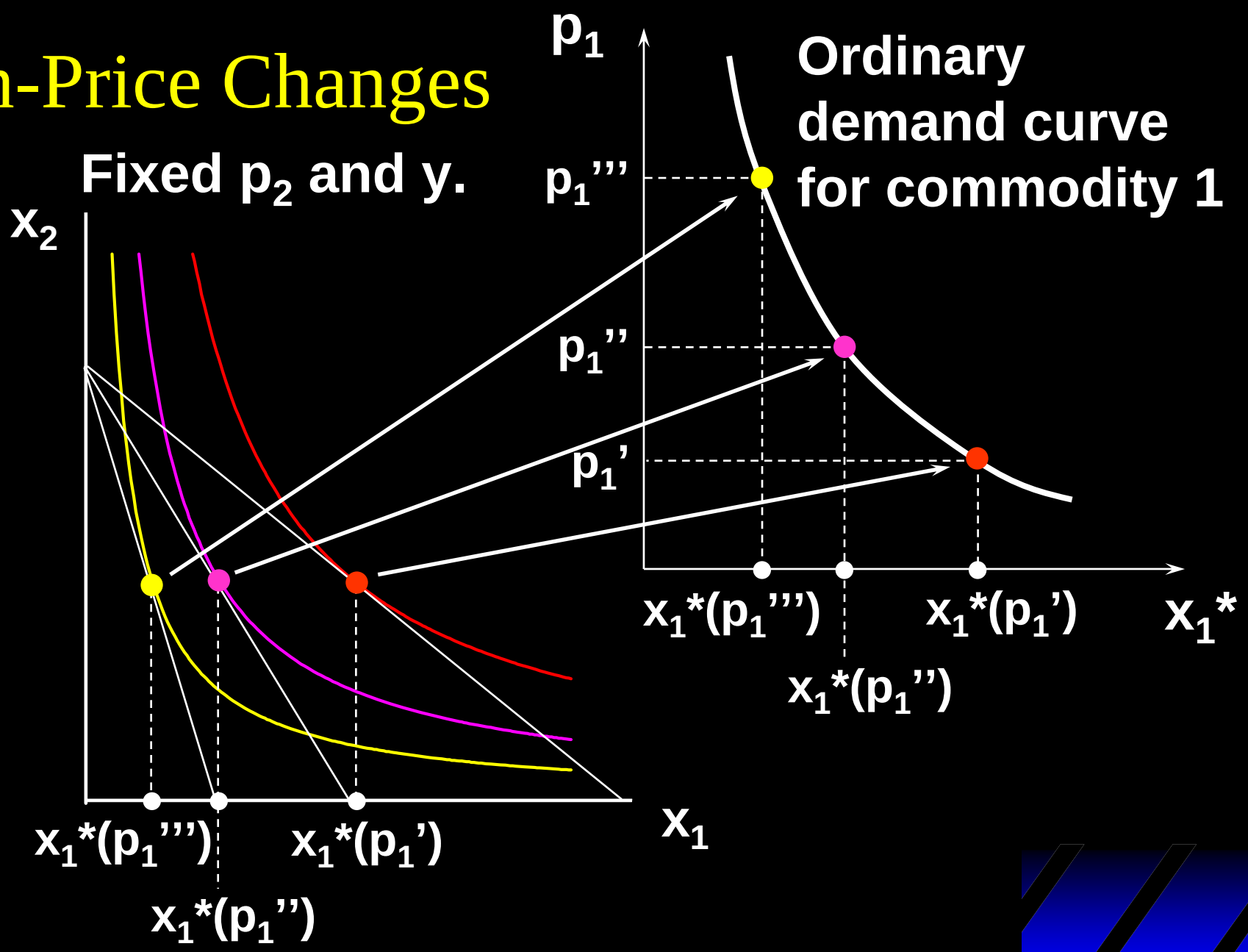
Own-Price Changes



Own-Price Changes

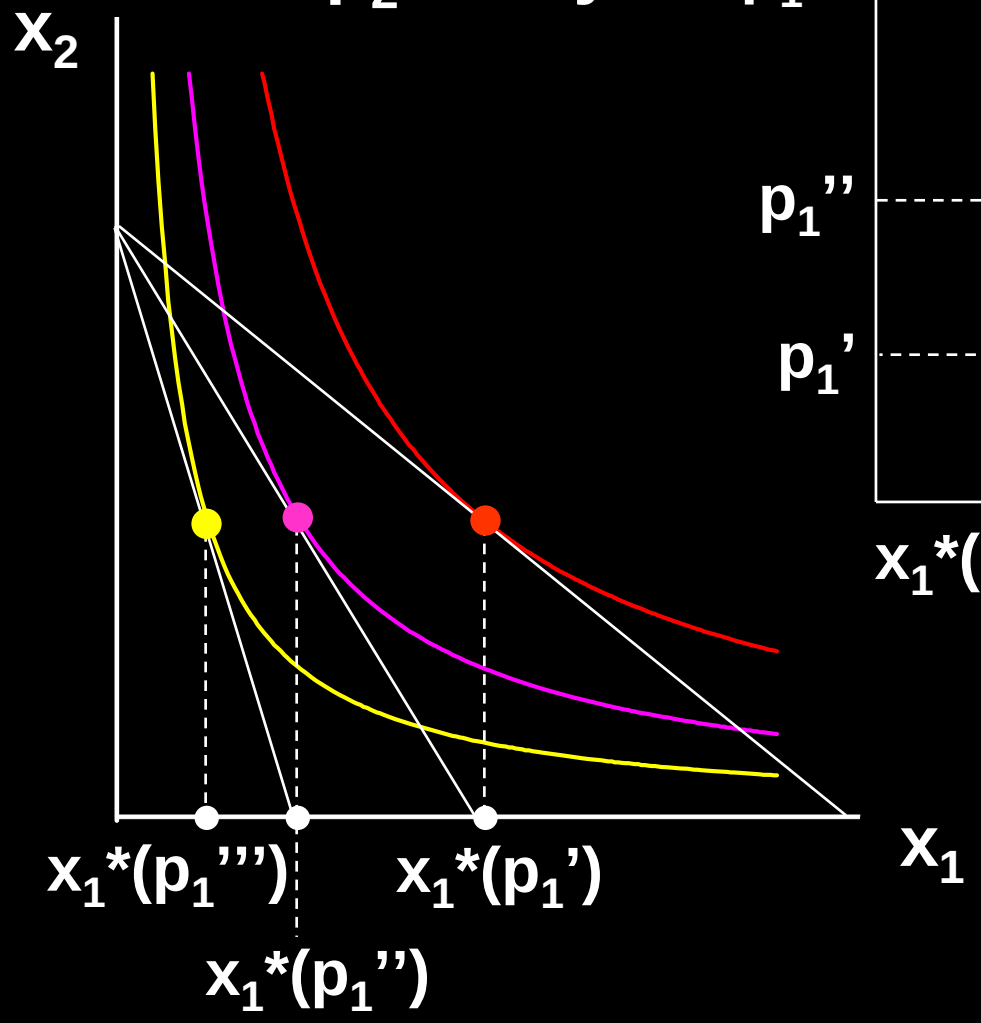


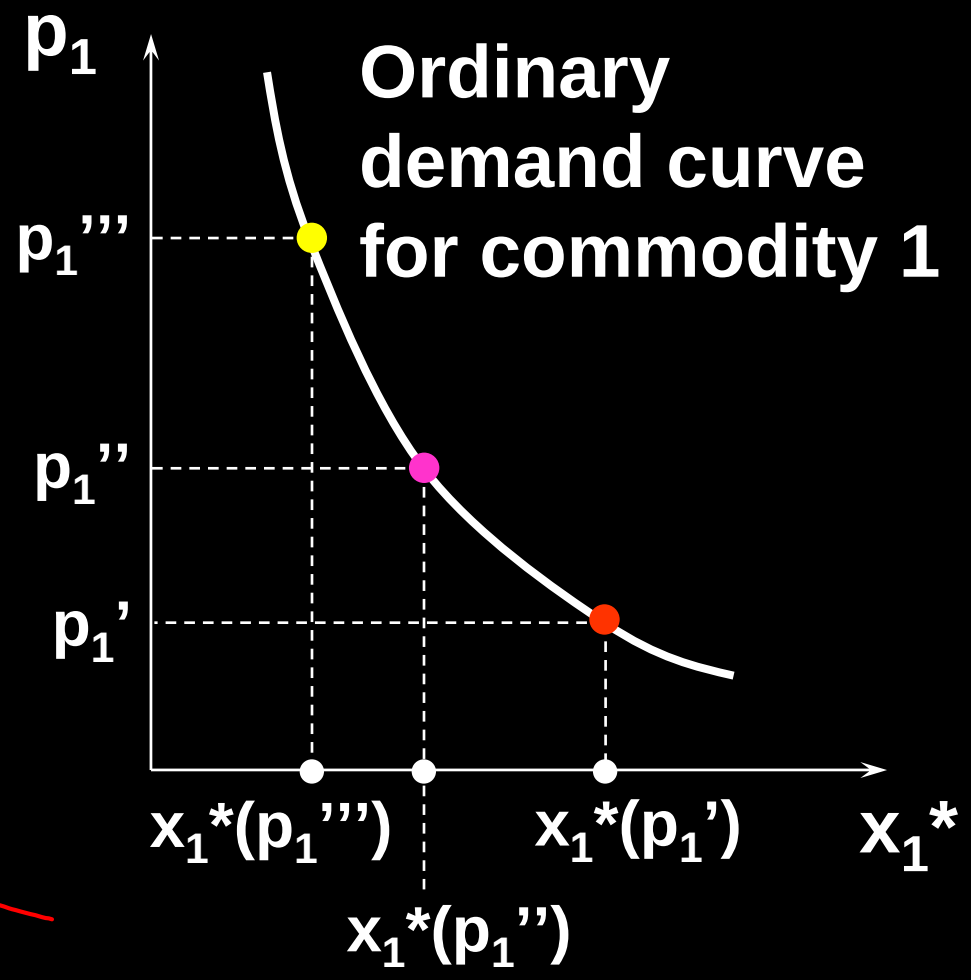
Own-Price Changes



Own-Price Changes

Fixed p_2 and y.





Own-Price Changes

- **What does ordinary demand look like for Cobb-Douglas preferences?**

Own-Price Changes

- What does ordinary demand look like for Cobb-Douglas preferences?
- Take

$$U(x_1, x_2) = x_1^a x_2^b.$$

Then the ordinary demand functions for commodities 1 and 2 are

Own-Price Changes

$$x_1^*(p_1, p_2, m) = \frac{a}{a+b} \times \frac{m}{p_1}$$

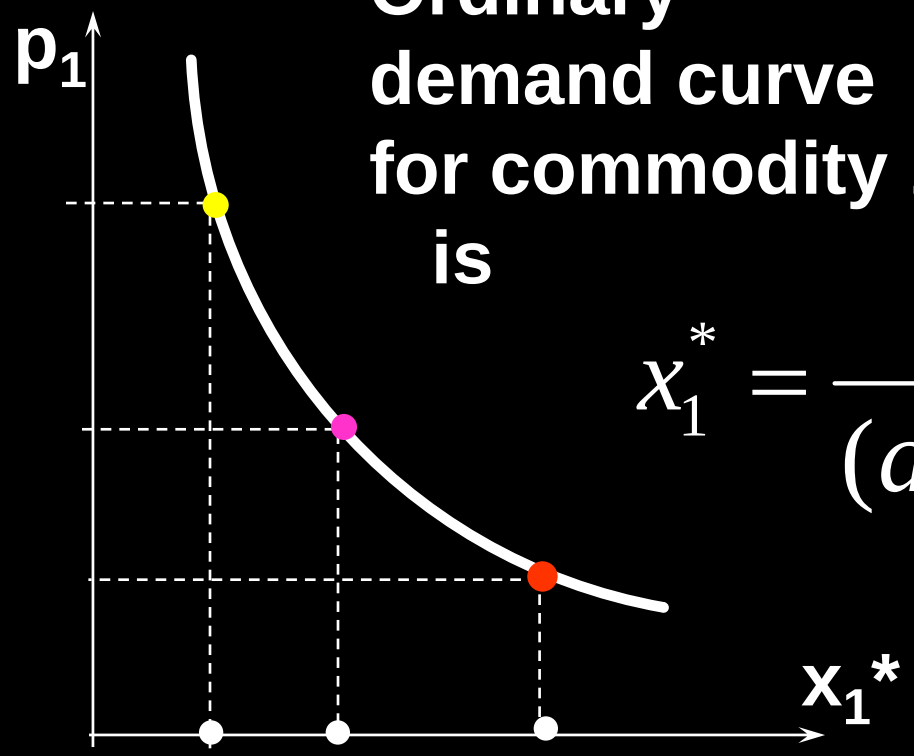
and

$$x_2^*(p_1, p_2, m) = \frac{b}{a+b} \times \frac{m}{p_2}.$$

Notice that x_2* does not vary with p_1

Own-Price Changes

Fixed p_2 and y.



Ordinary demand curve for commodity 1 is

$$x_1^* = \frac{am}{(a+b)\,p_1}$$

Own-Price Changes

- **The inverse demand function is the (ordinary) demand function viewing price as a function of quantity.**

Own-Price Changes

A Cobb-Douglas example:

$$x_1^* = \frac{am}{(a+b)p_1}$$

is the ordinary demand function and

$$p_1 = \frac{am}{(a+b)x_1^*}$$

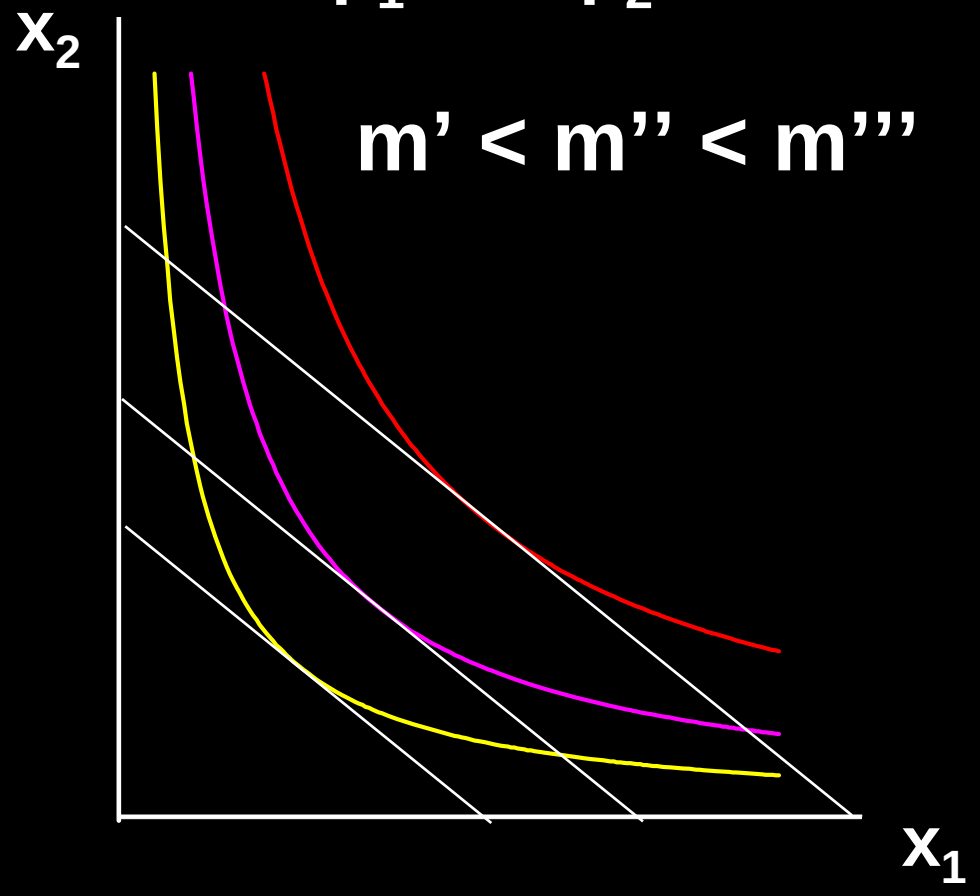
is the inverse demand function.

Income Changes

- **How does the value of $x_1^*(p_1,p_2,m)$ change as m changes, holding both p_1 and p_2 constant?**

Income Changes

Fixed p_1 and p_2.



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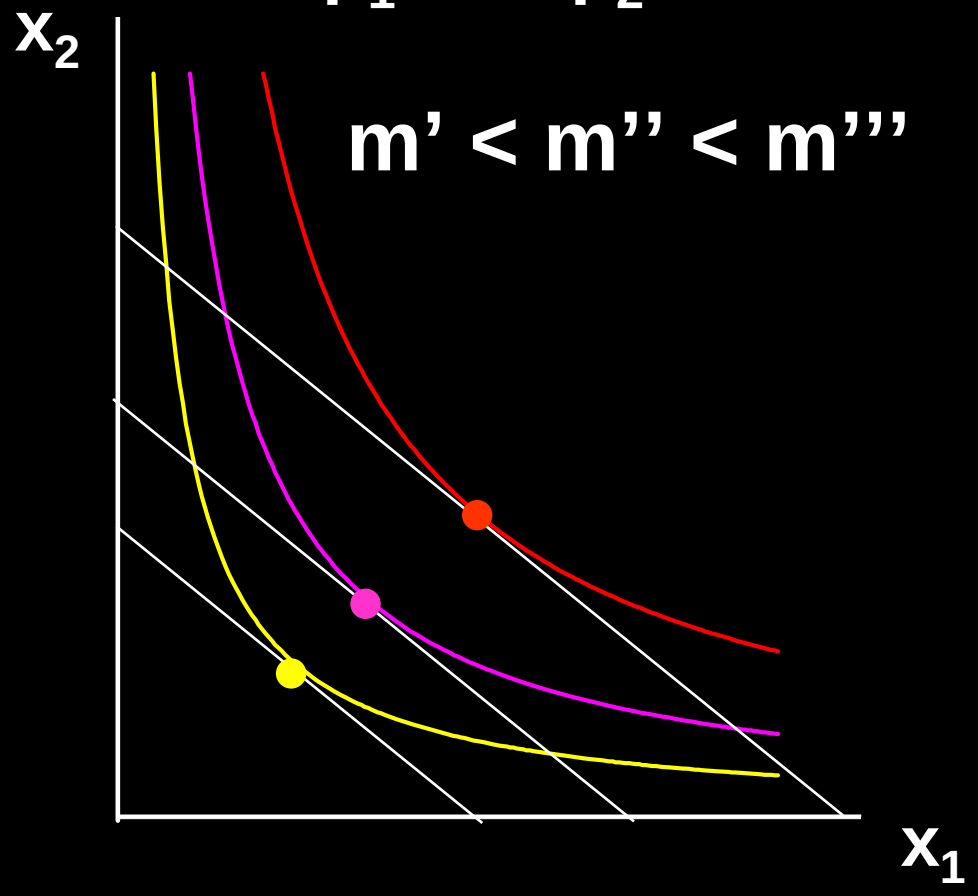

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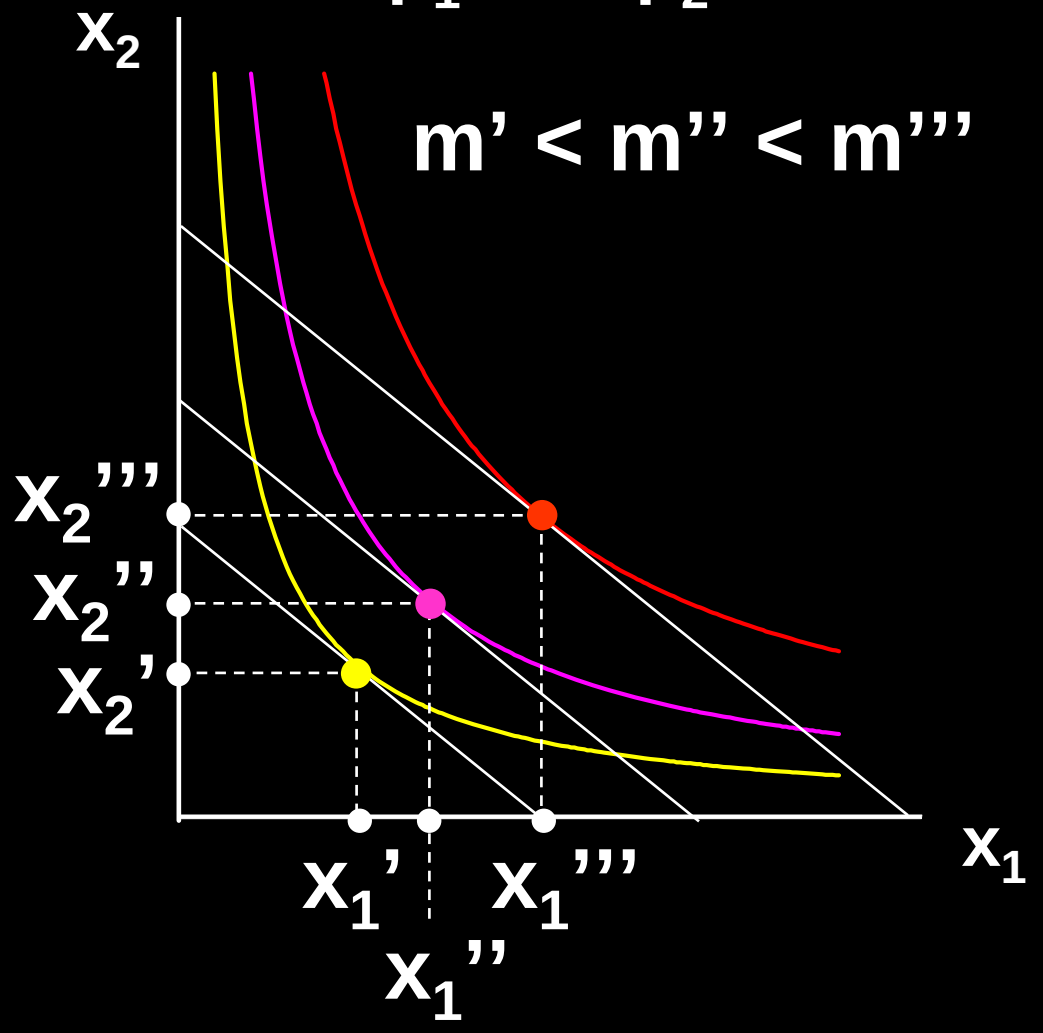


Income Changes

- **A plot of income as a function of quantity demanded is called an Engel curve.**

Income Changes and Cobb-Douglas Preferences

- **An example of computing the equations of Engel curves; the Cobb-Douglas case.**

$$U(x_1, x_2) = x_1^a x_2^b.$$

- **The ordinary demand equations are**

$$x_1^* = \frac{am}{(a+b)p_1}; x_2^* = \frac{bm}{(a+b)p_2}.$$

Income Changes and Cobb-Douglas Preferences

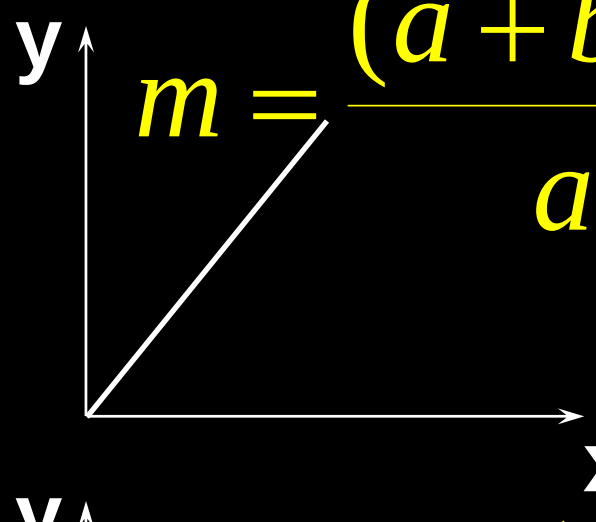
$$x_1^* = \frac{am}{(a+b)p_1}; x_2^* = \frac{bm}{(a+b)p_2}.$$

Rearranged to isolate y, these are:

$$m = \frac{(a+b)p_1}{a} x_1^*$$ **Engel curve for good 1**

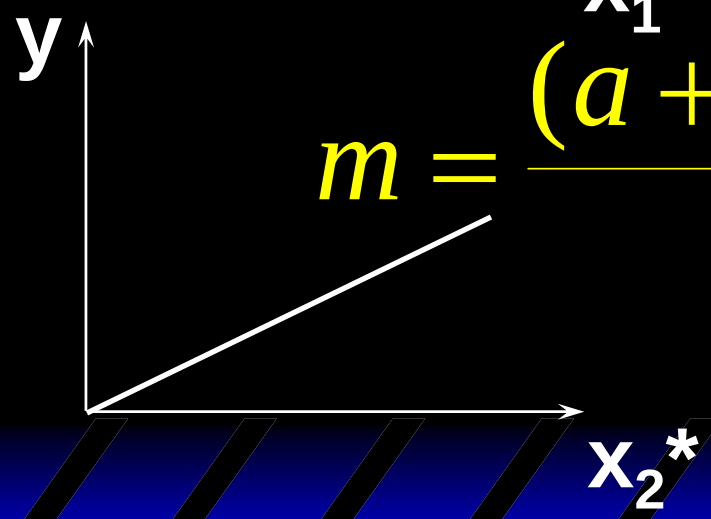
$$m = \frac{(a+b)p_2}{b} x_2^*$$ **Engel curve for good 2**

Income Changes and Cobb-Douglas Preferences



$$m = \frac{(a+b)p_1}{a} x_1^*$$

Engel curve for good 1



$$m = \frac{(a+b)p_2}{b} x_2^*$$

Engel curve for good 2

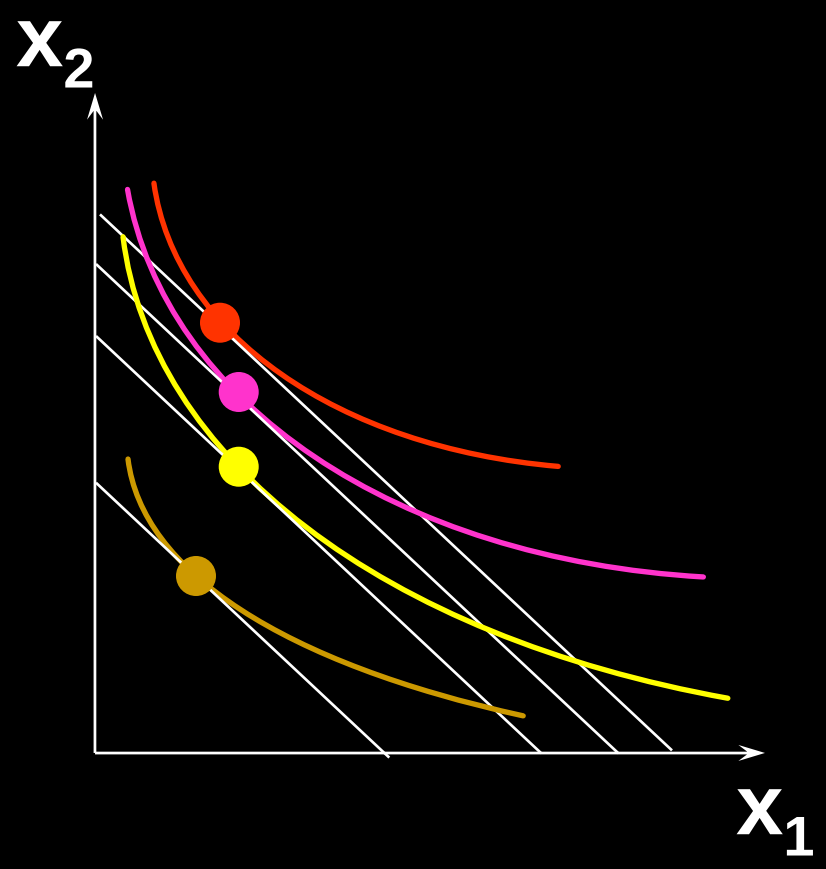
Income Effects

- **A good for which quantity demanded rises with income is called normal.**
- **Therefore a normal good's Engel curve is positively sloped.**

Income Effects

- **A good for which quantity demanded falls as income increases is called inferior.**
- **Therefore an income inferior good's Engel curve is negatively sloped.**

Income Changes; Good 2 Is Normal, Good 1 Becomes Income Inferior



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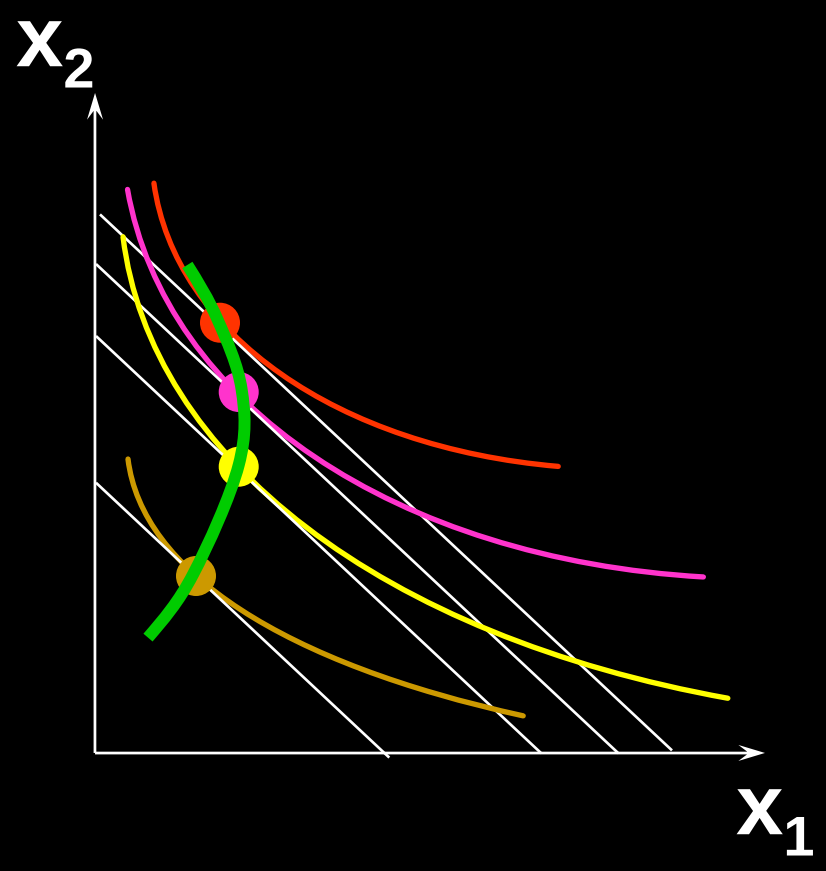

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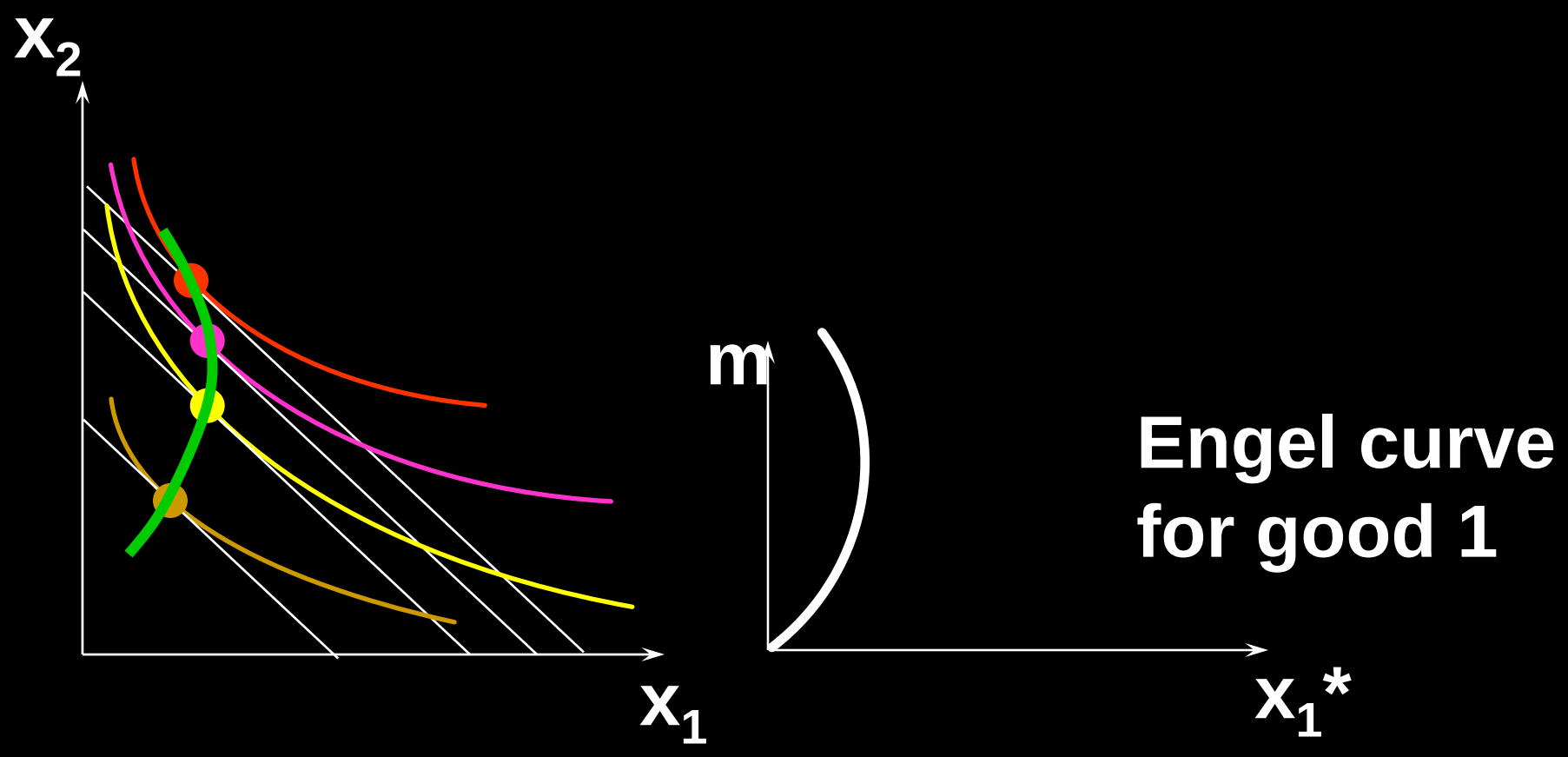
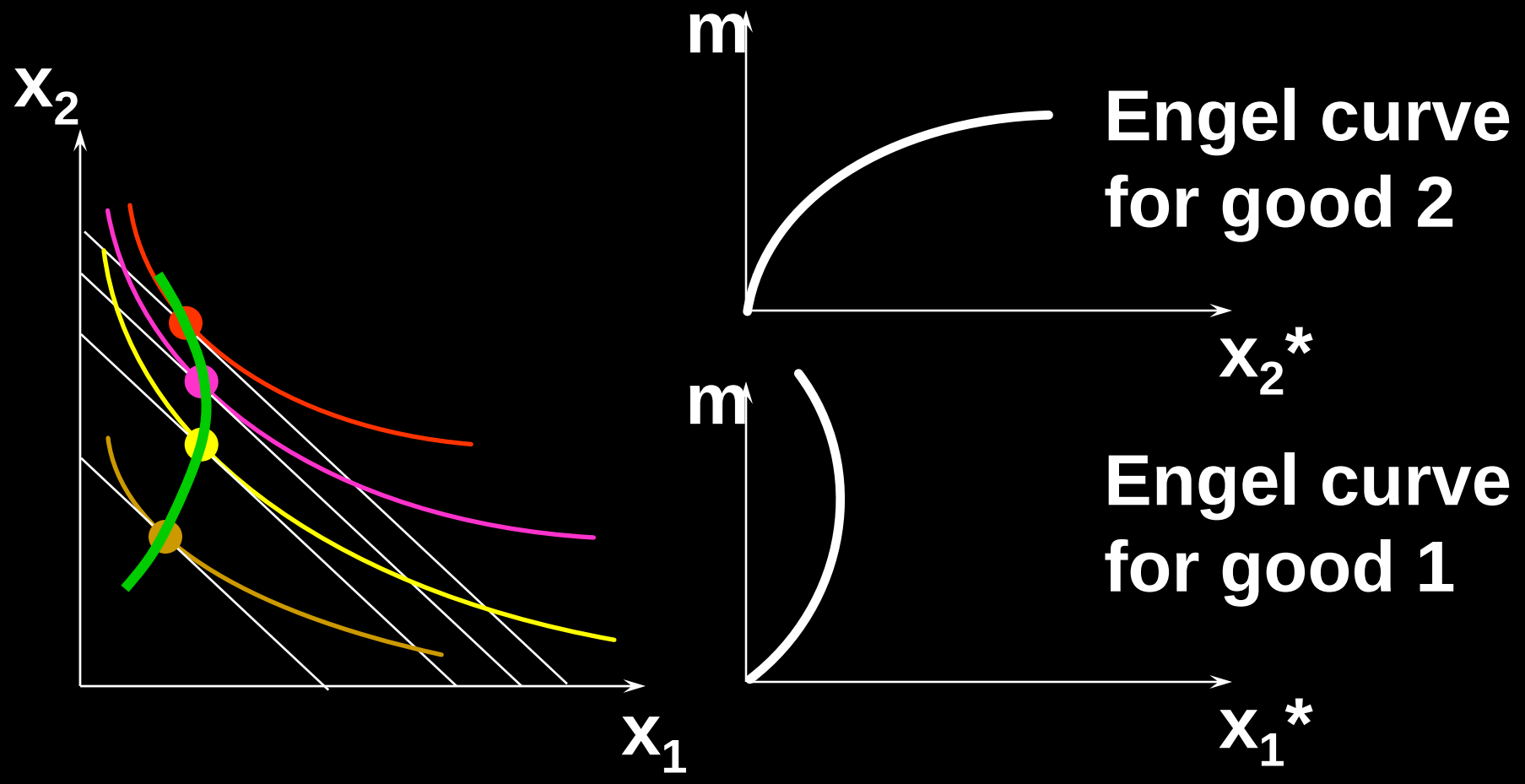


Income Changes; Good 2 Is Normal, Good 1 Becomes Income Inferior



Examples

- **Inferior goods:**

Examples

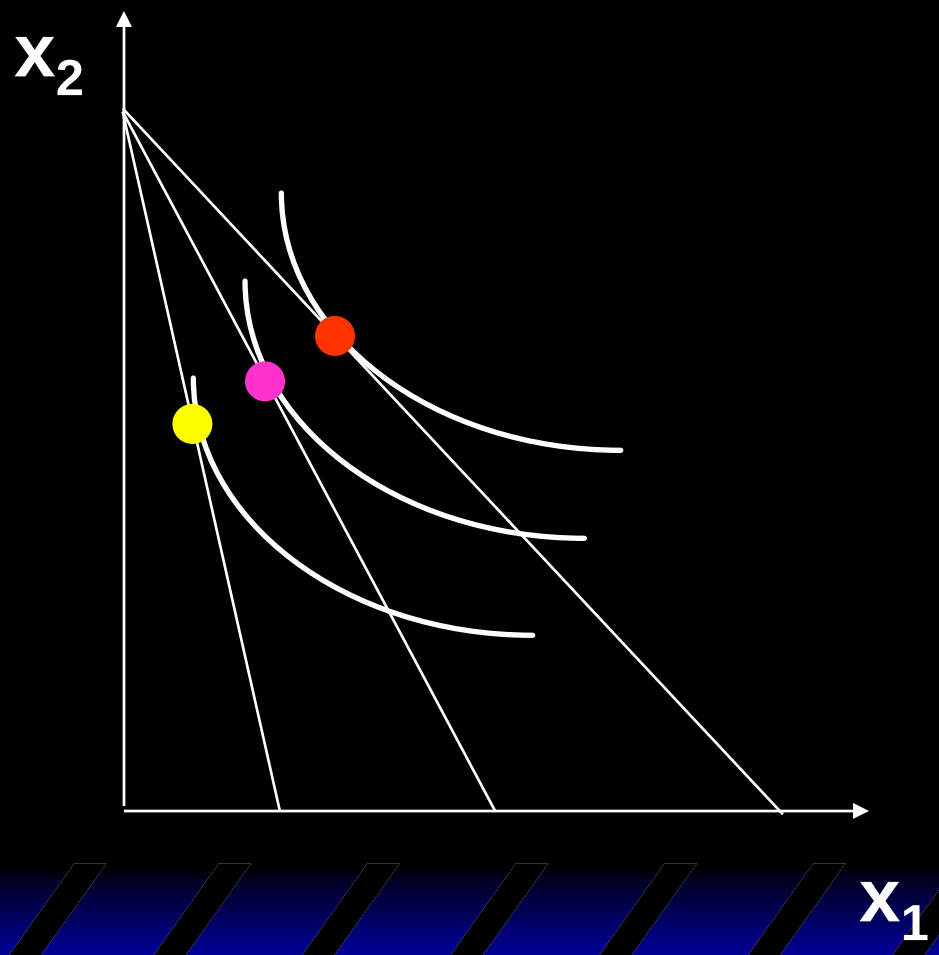
- **Inferior goods: Inner-city bus, hamburger, mass-market beer**
- **Potatoes**

Ordinary Goods

- **A good is called ordinary if the quantity demanded of it always increases as its own price decreases.**

Ordinary Goods

Fixed p_2 and m.



Ordinary Goods

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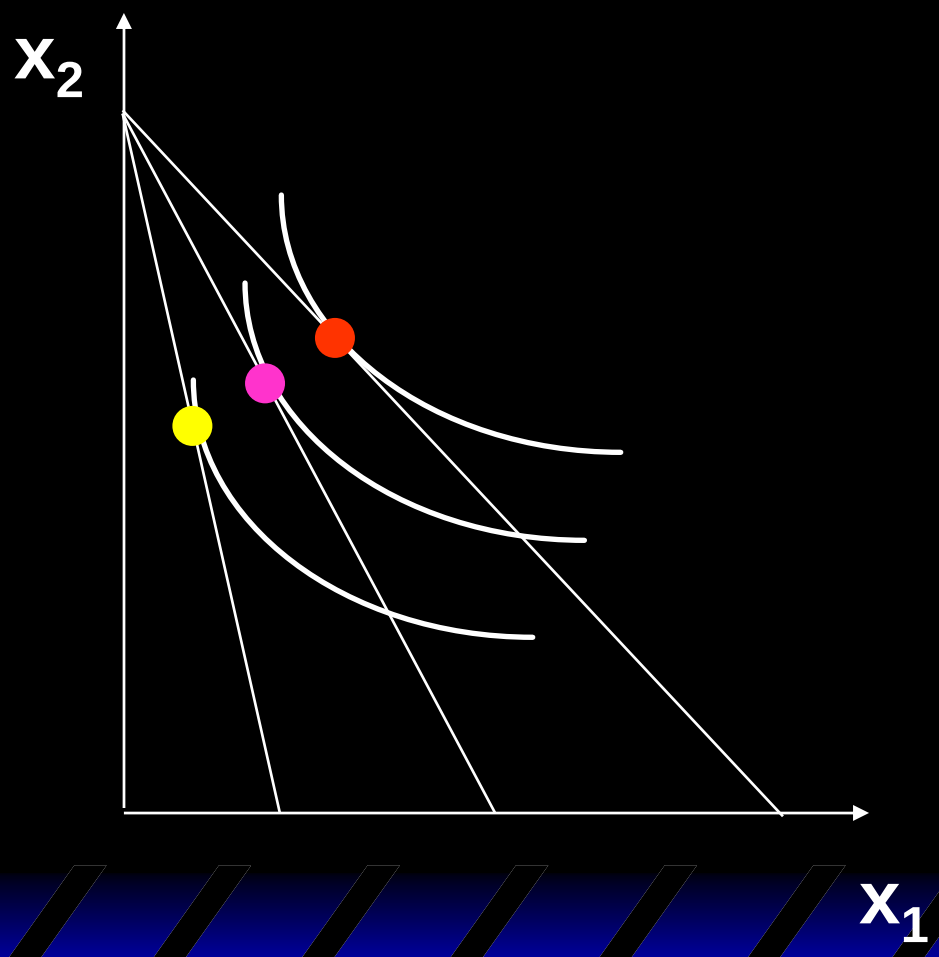
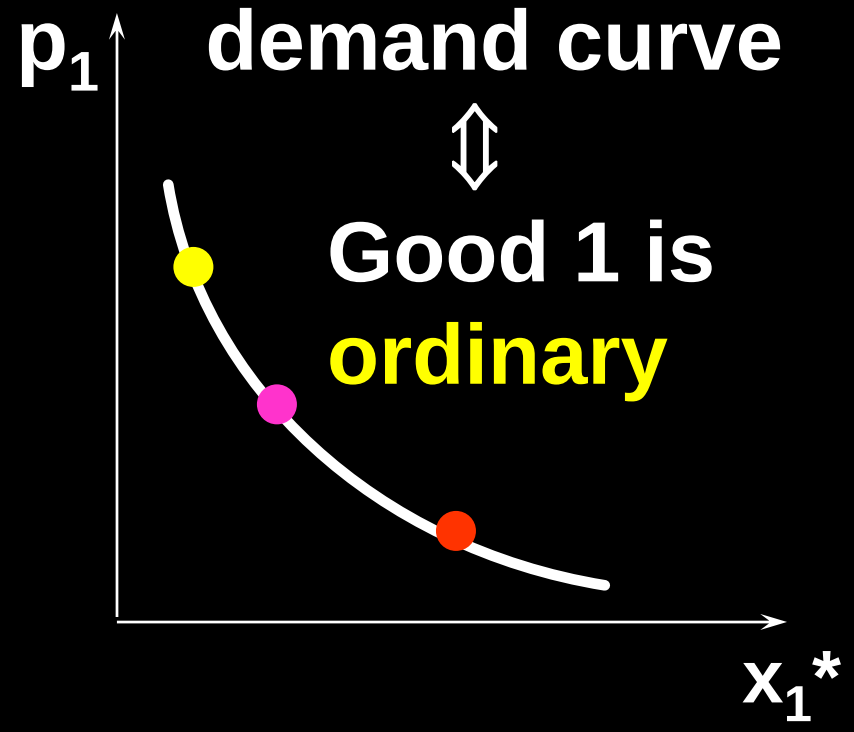


Downward-sloping demand curve

$\Updownarrow$

Good 1 is ordinary

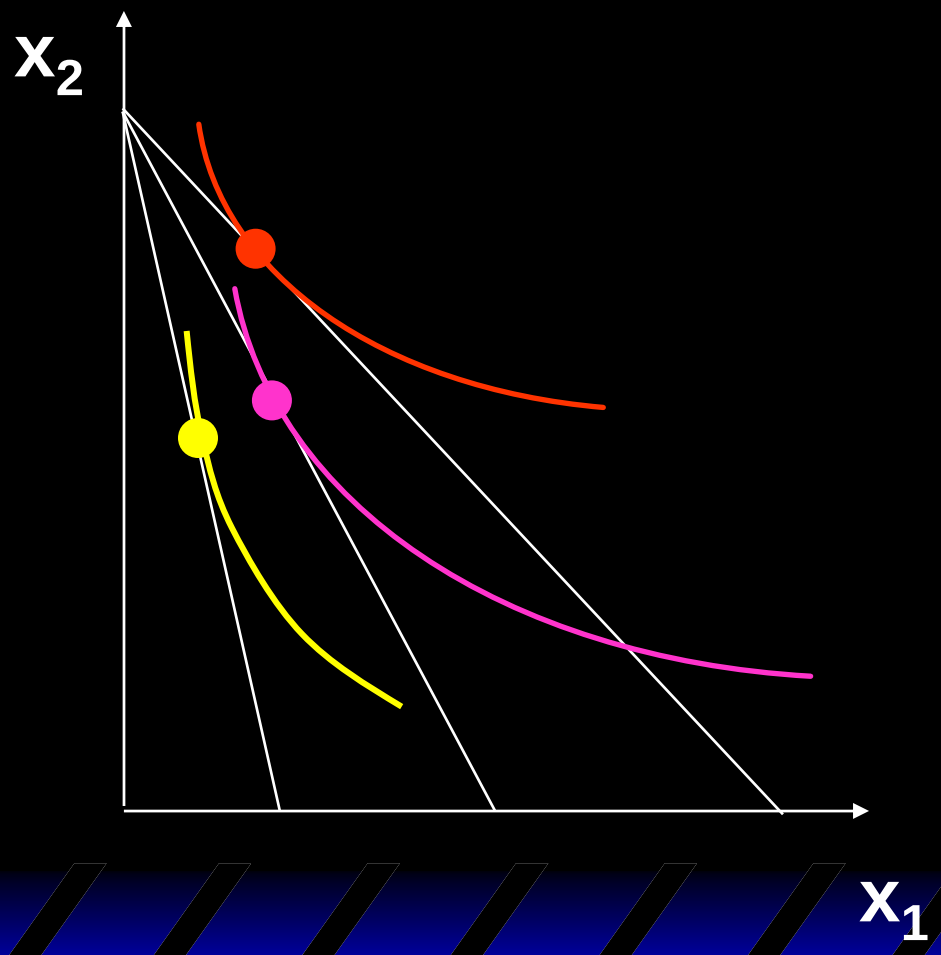


Giffen Goods

- **If, for some values of its own price, the quantity demanded of a good rises as its own-price increases then the good is called Giffen.**

Giffen Goods

Fixed p_2 and m.



Giffen Goods

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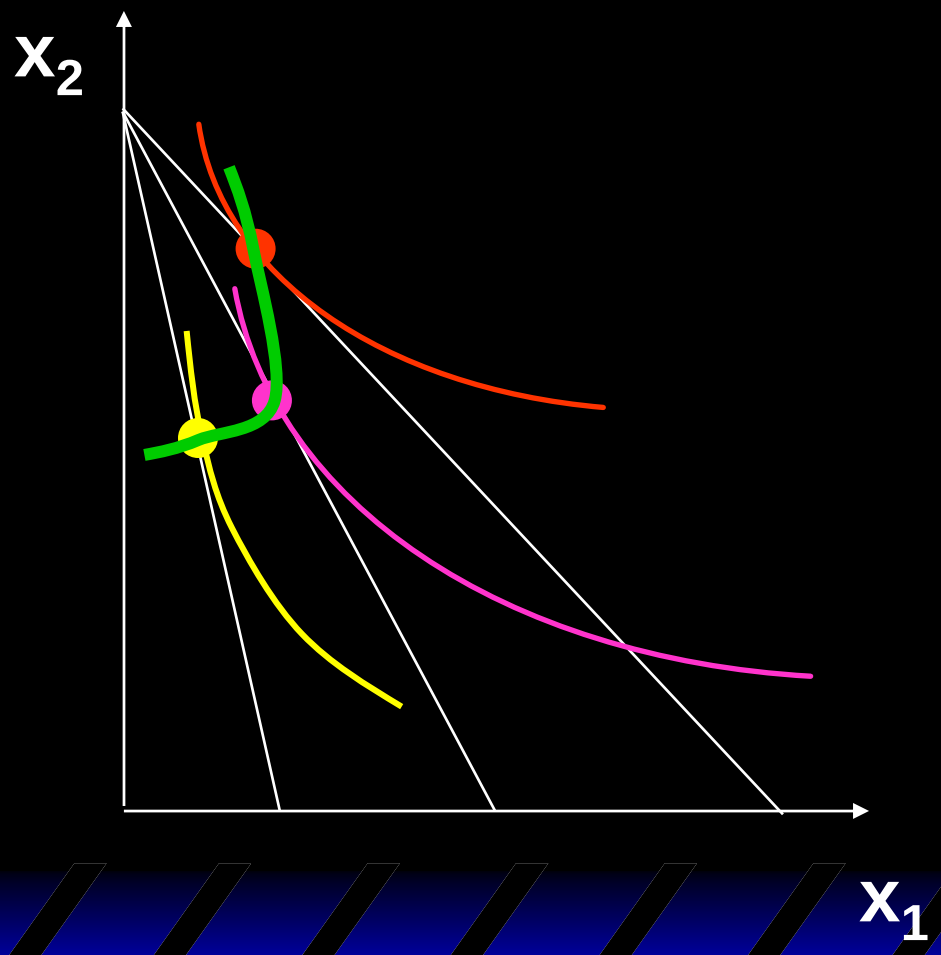

Giffen Goods

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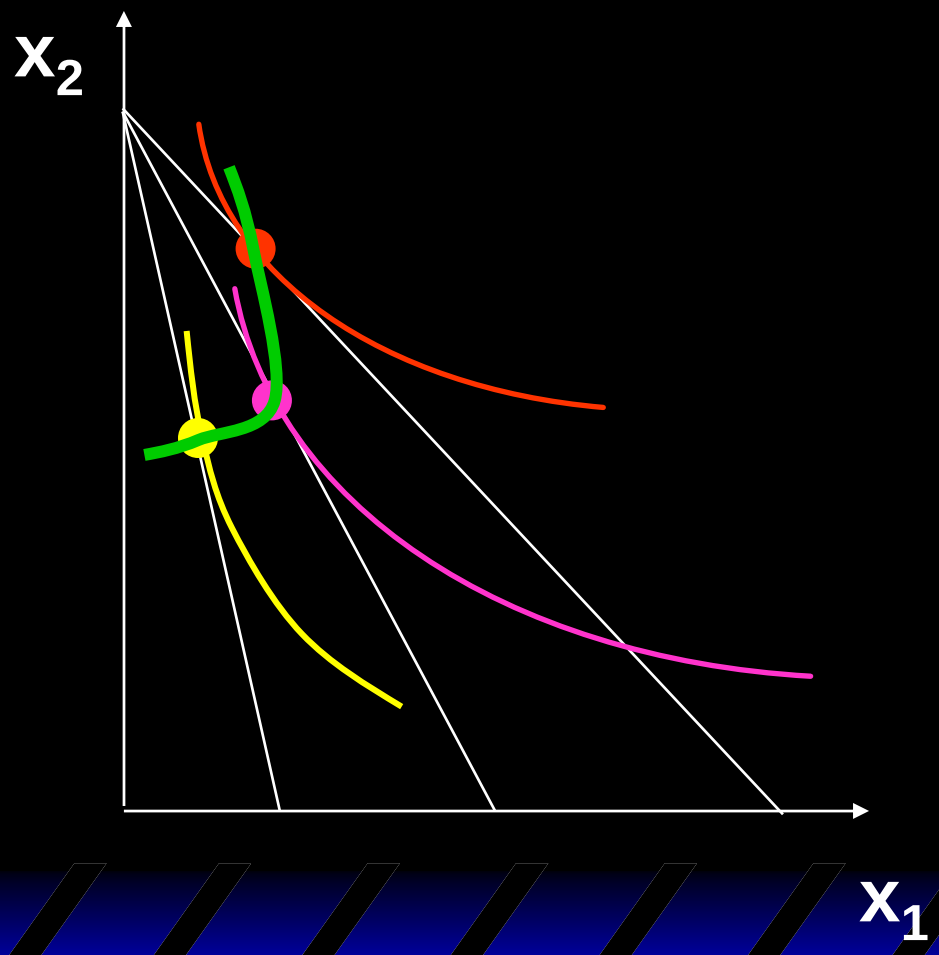

Demand curve has a positively sloped part

⇕

Good 1 is Giffen

Giffen Goods

- **"*As Mr. Giffen has pointed out, a rise in the price of bread makes so large a drain on the resources of the poorer labouring families… that they are forced to curtail their consumption of meat … [but] bread being the cheapest food… they consume more of it*." Marshall (1895).**

Cross-Price Effects

- **If an increase in p_2**
  - **increases demand for commodity 1 then commodity 1 is a gross substitute for commodity 2.**
  - **reduces demand for commodity 1 then commodity 1 is a gross complement for commodity 2.**

Cross-Price Effects

A perfect-complements example:

$$x_1^* = \frac{m}{p_1 + p_2}$$

so

$$\frac{\partial x_1^*}{\partial p_2} = -\frac{m}{(p_1 + p_2)^2} < 0.$$

Therefore commodity 2 is a gross complement for commodity 1.